

(TR-104) PYTHON WITH AI/ML

Training Day 72 Report

Date: 21 April 2026

Topic Covered: FastAPI Chatbot Backend Development, MongoDB Query Optimization & LangChain Agent Integration

The seventy-second day of training focused on developing the FastAPI backend for the MB Trailer chatbot project using MongoDB, LangChain agents, and Ollama models. The training also included query optimization, caching, asynchronous execution, and intelligent financial data analysis.

- Started development of FastAPI backend for chatbot functionality.
- Configured environment variables securely using dotenv.
- Established MongoDB connection with optimized settings including connection pooling and timeout handling.
- Worked with leasepayments collection for chatbot-based financial data analysis.
- Implemented MongoDB indexing on fields such as status, dueDate, and paymentMethod for faster query execution.
- Added caching mechanism with TTL support to optimize repeated database queries.
- Developed timezone-aware utility functions for proper date handling.
- Created `get_full_summary` tool for generating complete financial summaries including paid, pending, overdue, and total amounts.
- Implemented `get_payment_stats` tool for calculating counts and amounts based on payment status.
- Developed `get_payment_list` tool with filtering options including upcoming payments and result limits.
- Built `get_stats_by_method` tool for analyzing payment methods and statistics.
- Created `get_summary_by_entity` tool for trailer-wise and customer-wise payment summaries.
- Used `ThreadPoolExecutor` for parallel execution of aggregation queries to improve performance.

- Integrated LangChain agent with FastAPI backend for intelligent query handling.
- Configured Ollama model (qwen2.5:3b) for chatbot response generation.
- Defined structured system prompts with strict tool usage and response formatting rules.
- Developed /chat API endpoint for handling user chatbot queries.
- Implemented asynchronous execution using asyncio and run_in_executor for non-blocking operations.
- Added response extraction and formatting logic from agent outputs.
- Implemented proper exception handling for stable and reliable API responses.
- **GitHub Link:**
<https://github.com/JaspinderKaurWalia26/chatbot>

Overall, the training strengthened understanding of FastAPI backend development, MongoDB optimization, LangChain agents, asynchronous execution, intelligent chatbot systems, and AI-powered financial data processing.